Groups keep the amount - learner book

Draft • M01 • L-MATH-001-01-01 • tens and ones, 0-99

Use counters, groups of ten, paper and a pencil. Directions may be read aloud. You may speak, draw, point to objects, or ask an adult to write your explanation. A partner may be another learner or your teacher.

Our question

How can the groups change while the amount stays the same?

Say or show: one counter is our one-unit; one group of ten contains ten counters. Use the words one, ten, group, amount and equal.

Cycle A - Build and describe

Watch your teacher group fourteen counters. Name the grouped units. Explain what happened to the counters when a group was made.

A1 - Make groups

Start with sixteen counters. Make as many complete groups of ten as possible. Show the tens and the extra ones. Explain how you know no counters disappeared.

My model and explanation:

A2 - Listen and build

Your teacher will describe two collections. Build each one and label its tens and extra ones. Explain any zero position in your written numeral.

First collection:

Second collection:

A3 - Your partner has a clue

Keep your own card private. Ask your partner about the box whose clue you do not have. Build or draw the groups from the description; ask if your model agrees. Then swap roles. Show each complete collection with tens and extra ones.

Red box:

Blue box:

A4 - Show what you understand

Build or draw twenty-seven counters in complete tens and extra ones. Name the counted one-unit. Explain why making ten-groups keeps the amount.

My model and reason:

Cycle B - Exchange and explain

Watch your teacher unpack one ten from a collection of twenty-one counters. Ask: did any counters enter or leave the collection?

B1 - Unpack a ten

Build thirty-two with complete tens and extra ones. Unpack one ten. Show both stages. Explain why they have equal amounts.

Before: After:

My reason:

B2 - Test a claim

A learner says: “If I unpack one ten from four tens and three ones, I get three tens and three ones.”

Build or draw the change. Is the claim correct? Correct it and give a reason.

My correction and reason:

B3 - Reconstruct the other collection

Use Cycle B on your separate partner cards. Describe your collection without showing the card. Your partner reconstructs it, packs complete tens, and explains why the total stayed the same. Swap roles.

Green box in complete tens and extra ones:

Gold box in complete tens and extra ones:

Our explanation of the exchange:

B4 - Two ways, one amount

Show forty-six as complete tens and extra ones. Show it again after unpacking one ten. Explain to your partner why both models still show forty-six.

First model: Second model:

My explanation:

Think back

Show your partner one task you can explain. Name one task you want to try again. Your teacher will check the mathematics with you; confidence alone is not a result.

Groups keep the amount - workbook

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Practice only after the matching cycle. Use counters or drawings. An adult may read the directions and record your spoken explanation. Stop and ask for another demonstration when a group does not make sense.

After Cycle A

W1

Show twenty-eight with complete tens and extra ones. Label both units.

Model:

W2

Draw four tens and nine extra ones. Write the total numeral.

Model and numeral:

W3

Build seventy. Explain what its zero ones means.

Model and reason:

After Cycle B

W4

Build five tens and one extra one. Unpack one ten. Show both equal amounts.

Before: After:

W5

Start with two tens and fifteen ones. Pack complete tens. Name the complete tens and remaining ones.

Before: After:

W6

Someone says: “Ten ones can be exchanged for one one.” Correct the named unit and explain why it has the right value.

Correction and reason:

W7

Show thirty-eight in two ways: complete tens and extra ones, then after unpacking one ten. Label the units.

First: Second:

W8

Explain why the exchange in W5 changed the groups but kept the amount. Use a picture, objects or words another person can follow.

Explanation:

Hundreds and zero places

Grade 2 • draft • L-MATH-001-01-02 • PV3, 0-999

Use a blank place-value chart, objects, drawings and working paper. Directions may be read aloud; you may explain by speaking, drawing or pointing. Name the units, test the claim and explain so another person can follow.

P-PV3-02-01

Build or draw 406 and 230. Label hundreds, tens and ones. Explain each zero position.

Two models / labels / reasons:

P-PV3-02-02

Start with 140. Unpack one hundred into tens, then unpack one ten into ones. Show all three stages and explain why the amount stays the same.

Three models / reason:

P-PV3-02-03

Build 305 and 350. A person says they are the same because they use the same digits. Test the claim using the units.

Models / conclusion / reason:

P-PV3-02-04

Build 402. Explain its zero position, then show how ten tens can become one hundred without changing the amount.

Model / zero explanation / exchange:

Numerals, names and sums

Grade 2 • draft • L-MATH-001-01-03 • PV3, 0-999

Use a blank place-value chart, objects, drawings and working paper. Directions may be read aloud; you may explain by speaking, drawing or pointing. Name the units, test the claim and explain so another person can follow.

P-PV3-03-01

Write five hundred three as a numeral and sum of place values. Build or draw the same amount.

Model / numeral / sum:

P-PV3-03-02

Read 460. Write its name and expanded form. Explain the contribution of the 6 and the zero position.

Name / sum / explanation:

P-PV3-03-03

Test the claim 608 = 60 + 8. Build both amounts and explain your conclusion.

Models / conclusion / correction:

P-PV3-03-04

Write 340 in words and expanded form. Explain why each form names the same amount.

Name / sum / reason:

Counts and equal steps

Grade 2 • draft • L-MATH-001-01-04 • PV3, 0-999

Use a blank place-value chart, objects, drawings and working paper. Directions may be read aloud; you may explain by speaking, drawing or pointing. Name the units, test the claim and explain so another person can follow.

P-PV3-04-01

Give five terms by fives starting at 95. Include 95 as the first term. Explain the step.

Terms / step:

P-PV3-04-02

Count by ones from 198 through 202. Show the groups before and after 199 becomes 200. Separate adding one from exchanging equal-valued groups.

Terms / models / reason:

P-PV3-04-03

Give three terms by tens starting at 390. Include 390. Explain the boundary exchange.

Terms / exchange / reason:

P-PV3-04-04

Count by hundreds from 500 through 900. Explain what unit is added each time.

Terms / explanation:

Compare amounts and explain

Grade 2 • draft • L-MATH-001-01-05 • PV3, 0-999

Use a blank place-value chart, objects, drawings and working paper. Directions may be read aloud; you may explain by speaking, drawing or pointing. Name the units, test the claim and explain so another person can follow.

P-PV3-05-01

Put 204, 240 and 420 in increasing order. Explain the first differing place at each comparison.

Order / reasons:

P-PV3-05-02

Complete 708 __ 780 and 659 __ 599 using <, > or =. Read both comparisons and justify them.

Comparisons / reasons:

P-PV3-05-03

A person says 390 is greater than 409 because 9 is greater than 0. Test the claim using groups.

Models / correction / reason:

P-PV3-05-04

Compare 600 and 600. Explain the equality with a model or the place values.

Comparison / reason: